The AI Productivity Paradox

Explaining Within Income Group Heterogeneity in AI Driven Productivity Performance

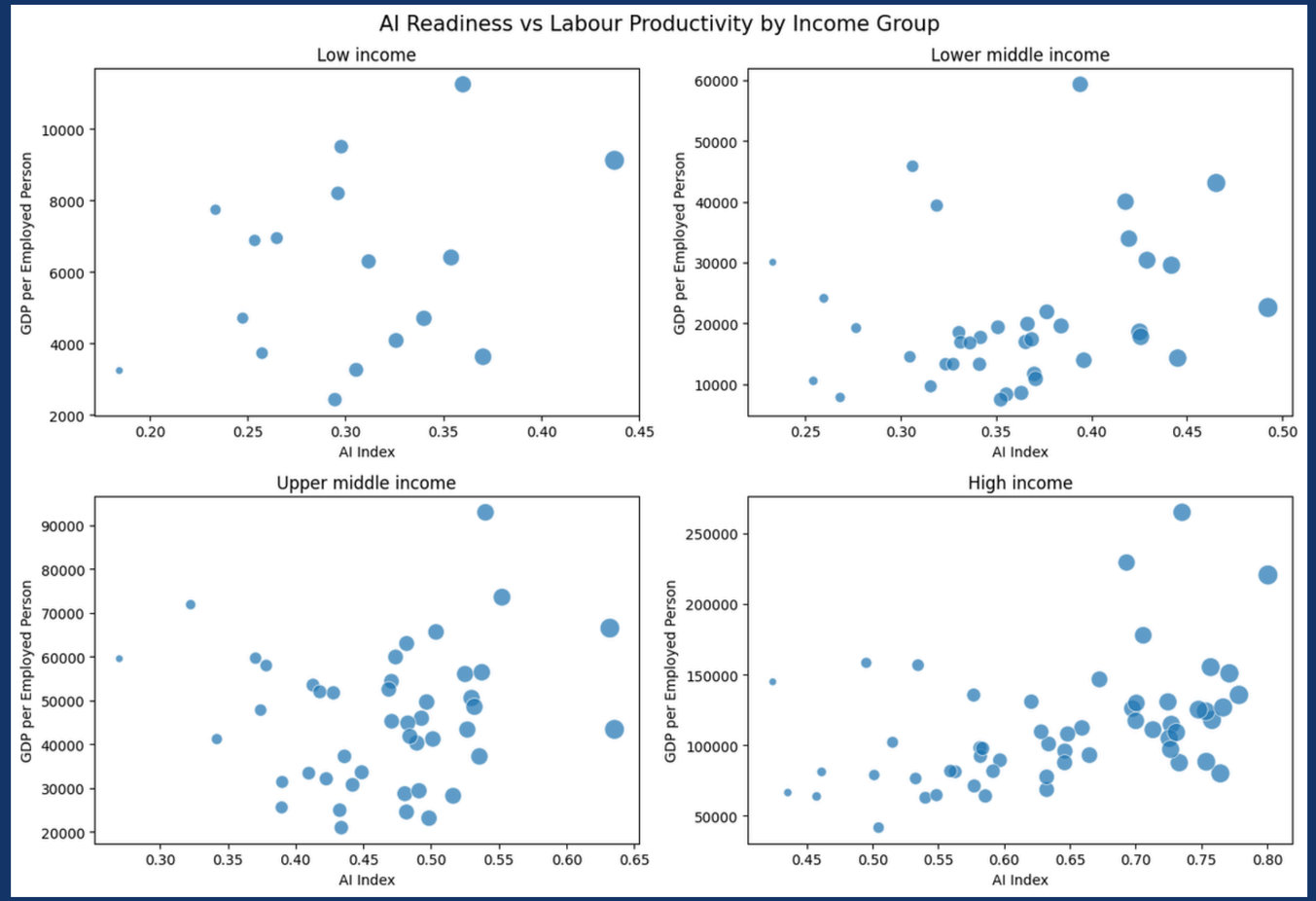


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Problem Statement

- Countries differ considerably in their preparedness for artificial intelligence, even within the same World Bank income group.
- Economic prosperity alone may not fully explain differences in countries' ability to develop and adopt AI.
- Factors such as human capital, digital infrastructure and government effectiveness may influence a country's readiness to benefit from AI.
- Access to digital technologies does not necessarily translate into higher productivity.
- Understanding these differences is important for identifying the factors associated with stronger AI readiness and economic performance.



Objective and Research Questions

To examine how economic, technological, and institutional factors are associated with differences in AI readiness among countries within the same World Bank income group.

1. Establishing Variation : Do countries within the same income group differ in their levels of AI readiness?
2. Explaining the Variation : How are differences in AI readiness associated with productivity, human capital, digital infrastructure and governance across countries, and what distinct country profiles emerge within each income group?

Dataset Description

Original variable	Meaning	Data source
AI Index	AI preparedness and readiness	IMF - AI Preparedness Index
hc	Human capital and workforce skills	Penn World Table (PWT)
rtfpna	Total factor productivity	Penn World Table (PWT)
Subscriptions	Fixed broadband subscriptions per 100 people	World Bank WDI / ITU
GDP_emp	GDP per employed person (labour productivity)	World Bank WDI
Internet	Internet users (% of population)	World Bank WDI / ITU
RD	Research and development expenditure or activity	World Bank WDI
Service share	Services value added (% of GDP)	World Bank WDI
Industry share	Industry value added (% of GDP)	World Bank WDI
Mobile connectivity	Mobile cellular subscriptions per 100 people	World Bank WDI / ITU
Government Effectiveness	Quality of public services and policy implementation	World Bank WGI

156 countries data

New Variable Constructed:
categorises countries into four World Bank income groups: Low, Lower-middle, Upper-middle and High income.

Project Tools & Architecture



Libraries: Pandas, NumPy, Statsmodel, Scikit-learn, SciPy, Matplotlib & Seaborn.

Data Cleaning and Preprocessing

1 Data merging

What was done
Joined AI Index, Penn World Table (hc, rtfpna), WDI and WGI on country. Renamed countries to match spellings. Libya, PNG and Timor-Leste had no PWT match and were lost.

Why it matters
Each source spells country names differently, so unmatched rows silently drop out.

2 Initial cleaning

What was done
Converted "no data" and ".." to NaN, removed 13 regional aggregates (G7, Euro area etc.) and converted text columns to numbers. 156countries remained.

Why it matters
Python can only calculate on true numbers, and the study needs real countries, not groups.

3 Income grouping

What was done
Mapped every country to a World Bank income group: High , Upper-middle , Lower-middle , Low .

Why it matters
The topic compares countries within the same income group, so each needs a label.

4 Missing values

What was done
Dropped R&D (51% missing). Filled other gaps with the income-group median (if skewed) or mean (if balanced).

Why it matters
Too much missing data cannot be filled reliably. Group-wise filling keeps the sample size.

5 Log transformation

What was done
Logged GDP per employed person (skew 1.29 to −0.60) and Mobile connectivity (skew 2.16 to −0.11).

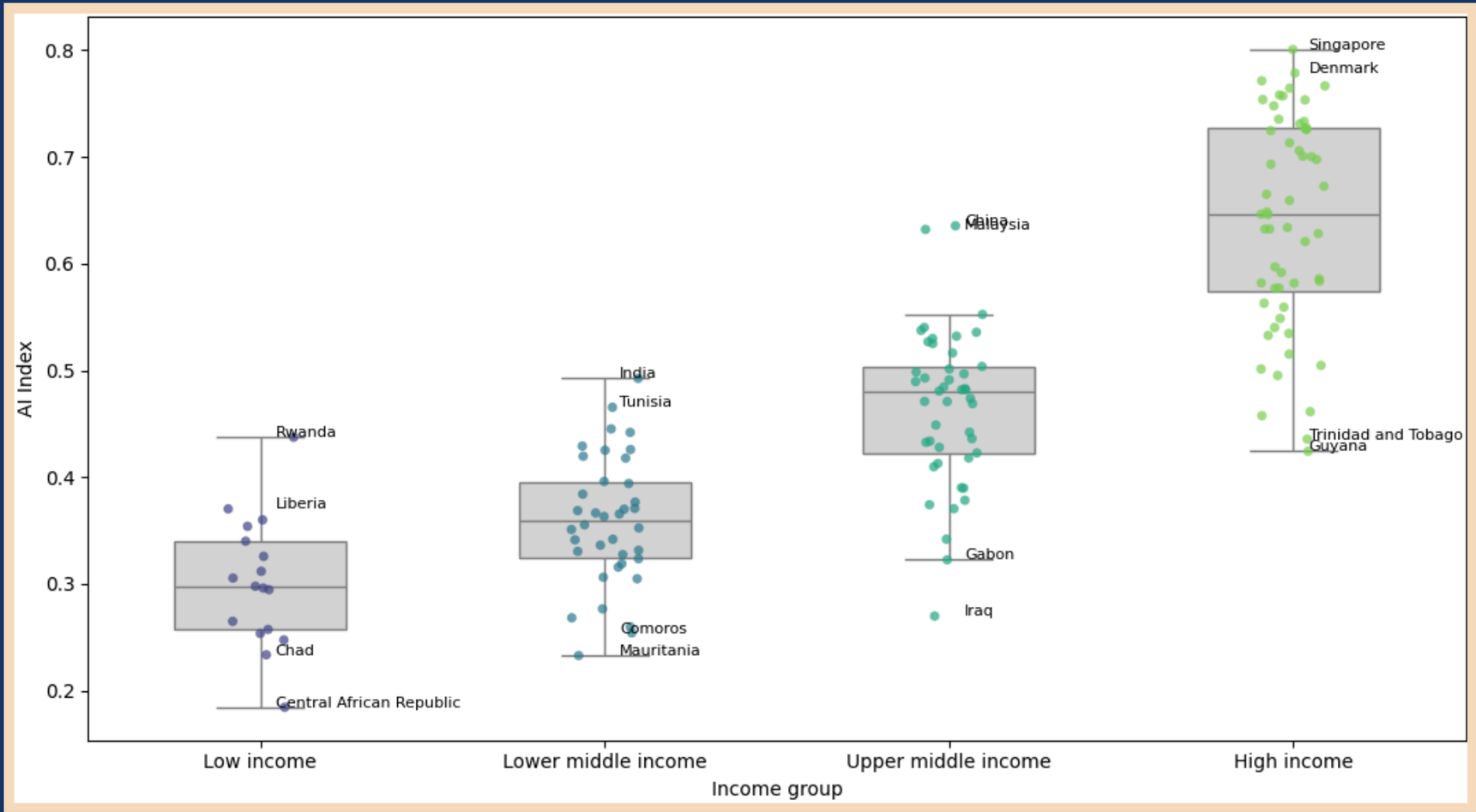
Why it matters
A few very rich or connected countries stretch the data. Logs balance it for regression.

6 Variable selection and checks

What was done
Dropped Industry share and rtfpna (near-zero correlation with AI Index). VIF under 3 within income groups. Kept outliers China, Malaysia and Iraq.

Why it matters
Weak variables add noise. Low VIF means no overlap. Outliers are the differences we want to explain.

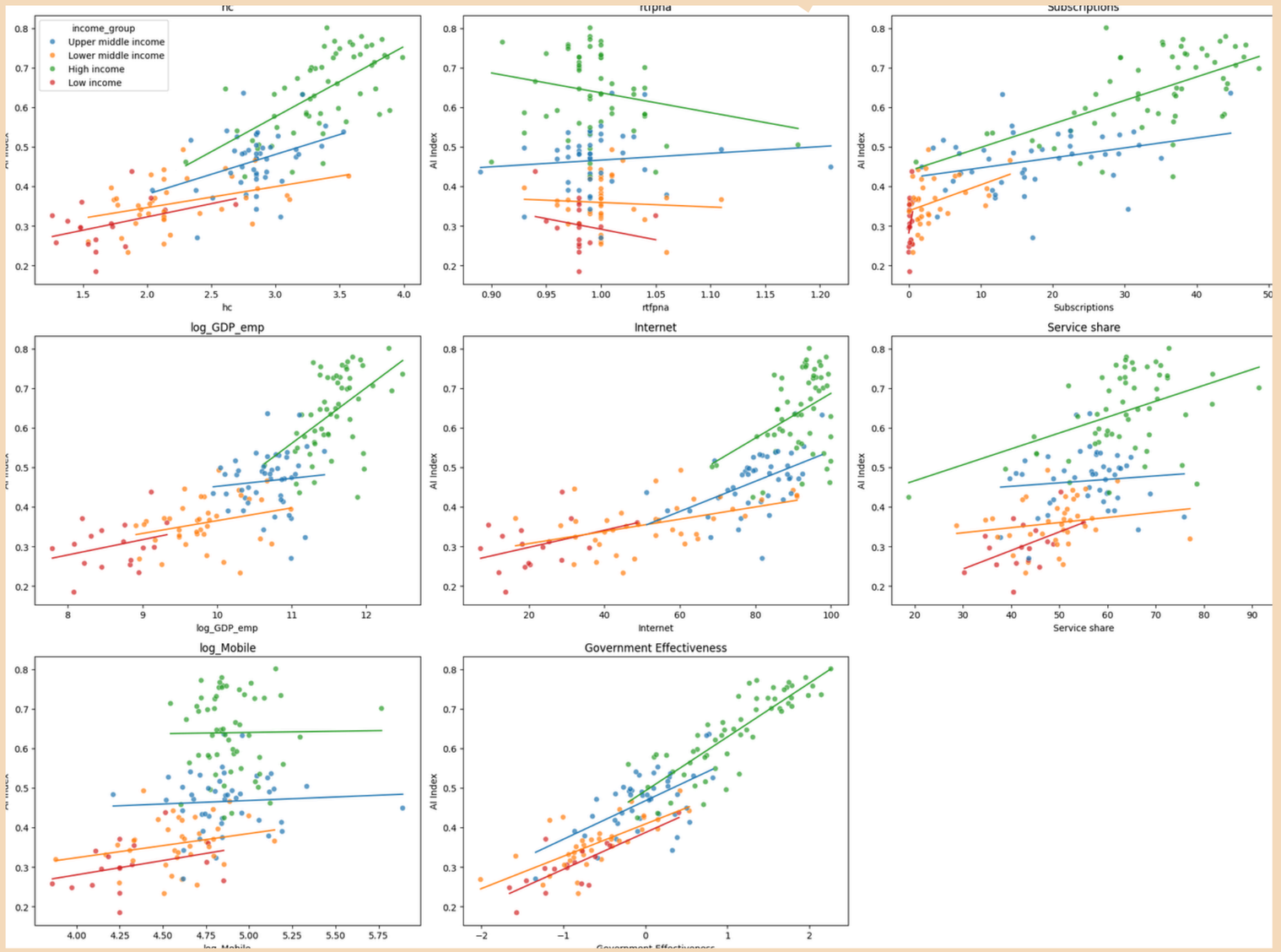
Exploratory Data Analysis



AI readiness generally increases with income level

There is substantial variation within income groups.

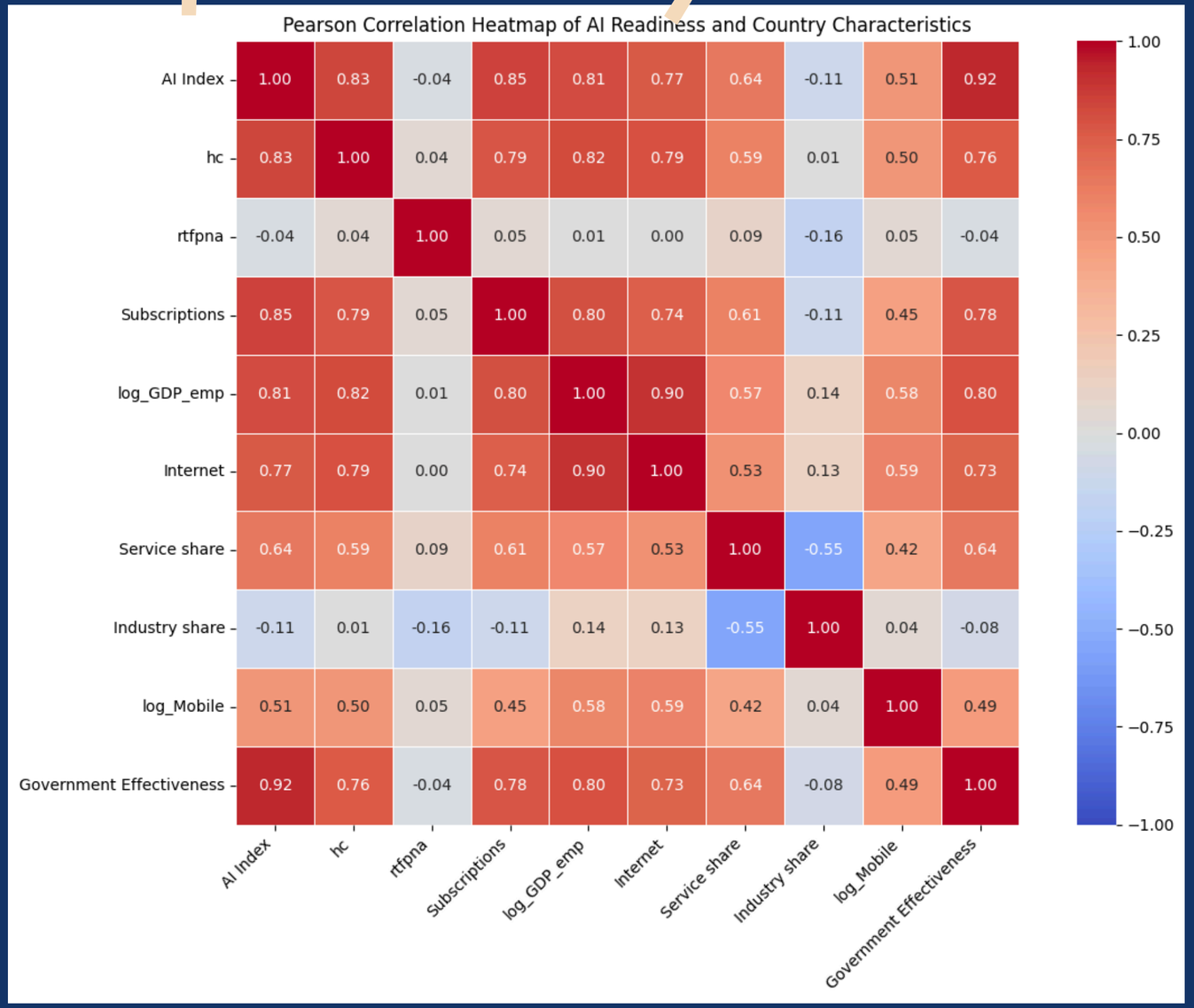
Exploratory Data Analysis



Government effectiveness, human capital and internet access show the clearest positive visual relationships with AI readiness. Mobile connectivity and TFP display less consistent patterns across income groups.

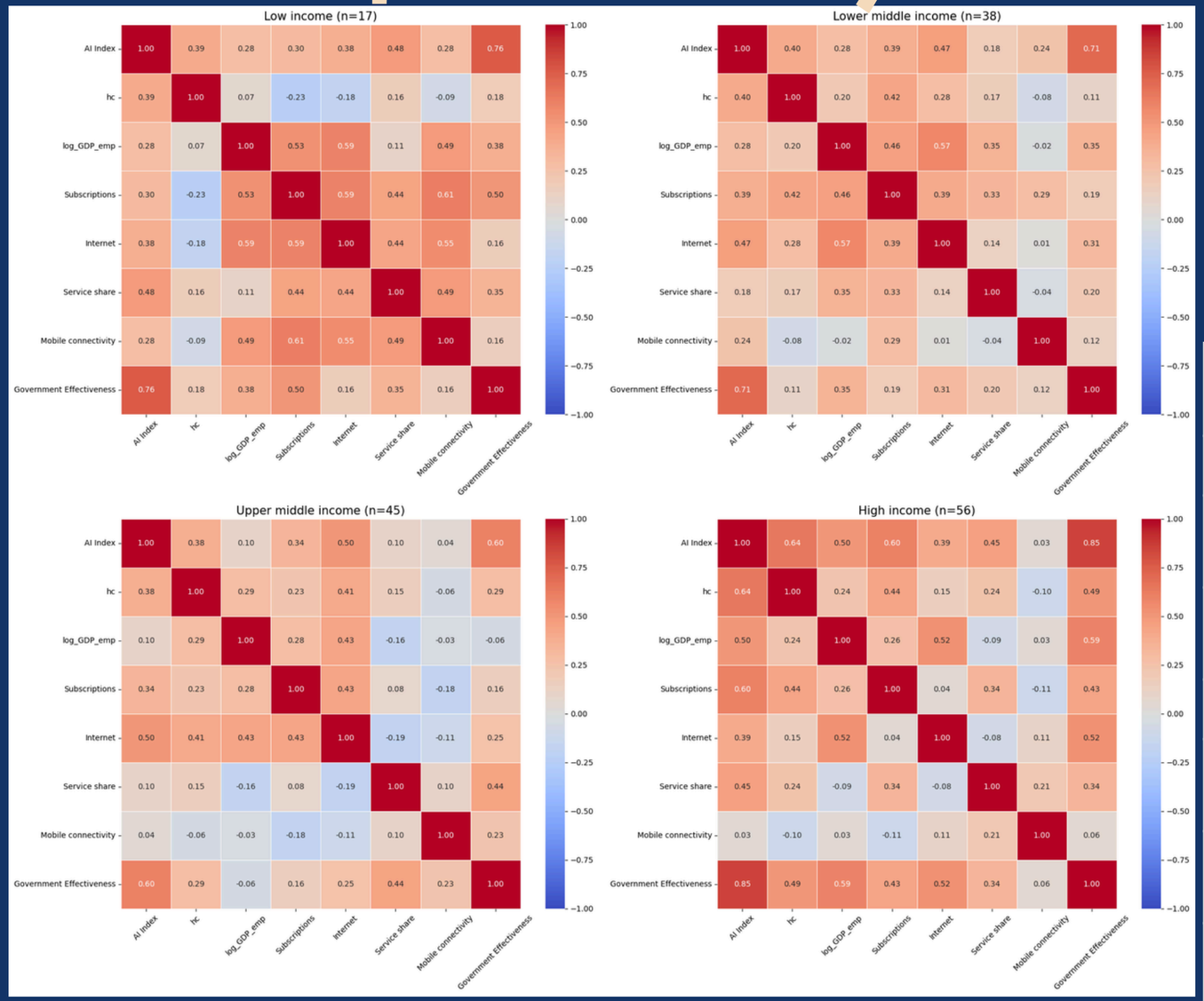
Relationship strength	Variables
Strong	Government Effectiveness, Human Capital, Log GDP per employed person
Moderate	Internet Access, Subscriptions
Weak	Service Share, Mobile Connectivity, TFP

Exploratory Data Analysis



AI readiness has the strongest positive correlation with government effectiveness(0.92), followed by subscriptions(0.85), human capital(0.83) and labour productivity(0.81).

Exploratory Data Analysis



Within each group, correlations between explanatory variables are much weaker, so the high pooled correlations come from rich-vs-poor differences. → Income group as controls

Variable	All countries	Low income	Lower middle	Upper middle	High income
hc	3.93	1.49	1.29	1.35	1.48
log_GDP_emp	7.74	2.61	1.9	1.35	1.95
Subscriptions	3.74	2.93	1.93	1.3	1.53
Internet	5.72	2.6	1.64	1.92	1.71
Service share	1.84	2.13	1.29	1.5	1.54
log_Mobile	1.61	2.05	1.28	1.11	1.12
Government Effectiveness	3.61	1.85	1.2	1.62	2.88

Model Building

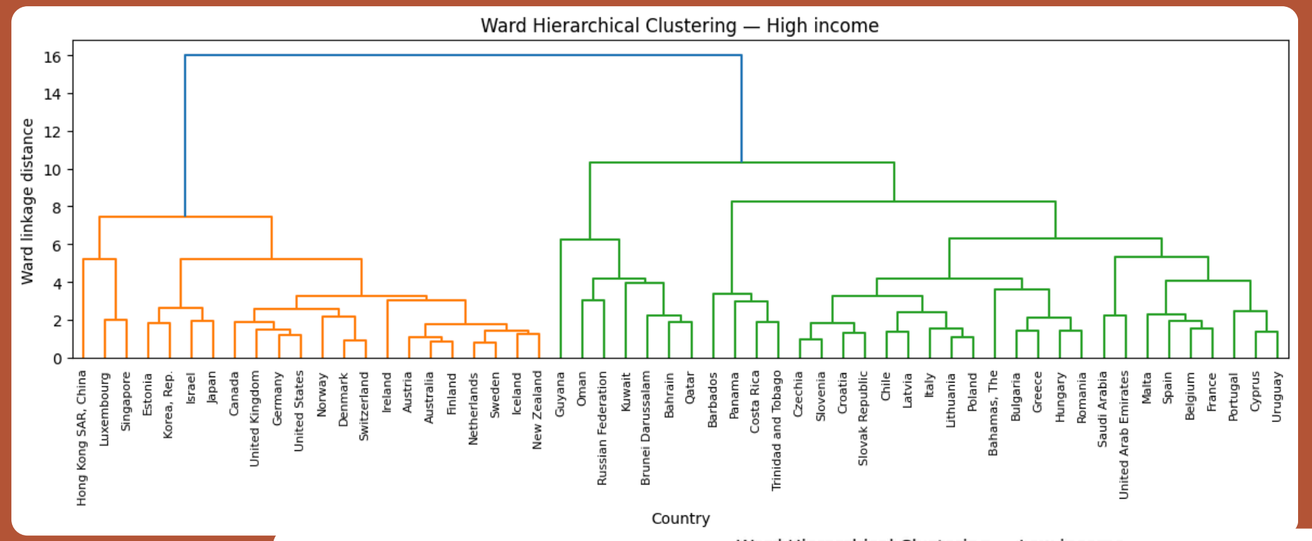
- Which factors are associated with differences in AI readiness across countries
- Controlling for Income Groups
- R^2 = 0.915
- human capital, broadband subscriptions, internet usage and government effectiveness have statistically significant positive associations with AI readiness.
- Wald Test :
  Null - All three coefficients are zero
  Alternative - Atleast one coefficient is non-zero
- Model 2 with interaction dummies
- Insignificant Values

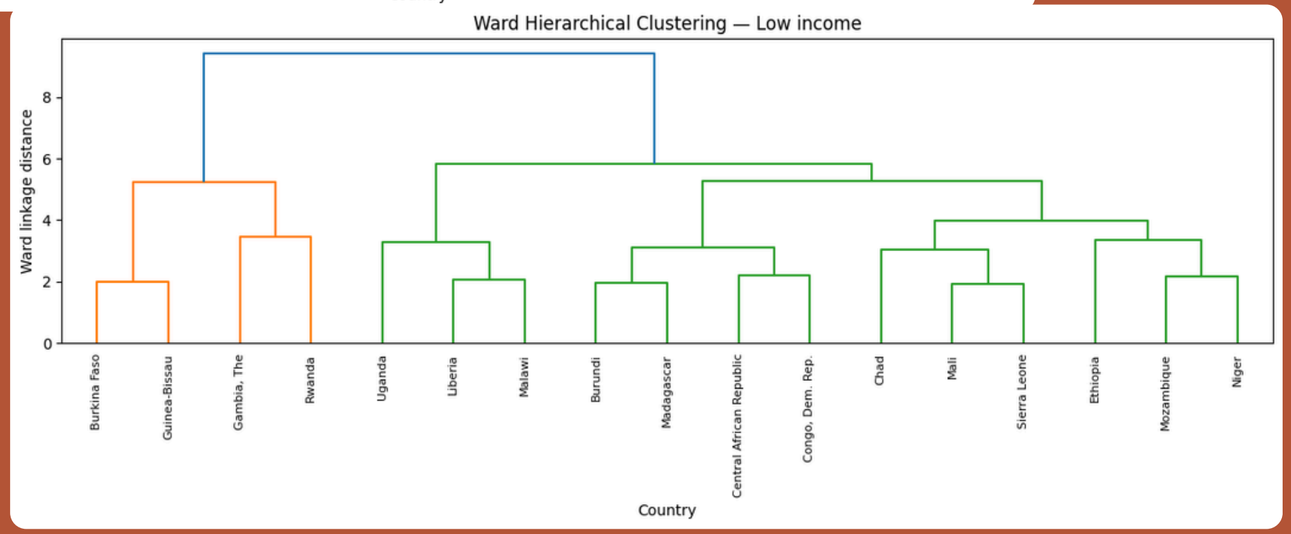
```
formula_full = (
    'Q("AI Index") ~ hc + Subscriptions + '
    'log_GDP_emp + Internet + log_Mobile + '
    'Q("Service share") + '
    'Q("Government Effectiveness") + '
    'C(income_group)'
)
```

```
R-squared: 0.9149723027254528
Adjusted R-squared: 0.9091083236030703
                            OLS Regression Results
==============================================================================
Dep. Variable:          Q("AI Index")   R-squared:                       0.915
Model:                            OLS   Adj. R-squared:                  0.909
Method:                 Least Squares   F-statistic:                     210.9
Date:                Mon, 26 Oct 2026   Prob (F-statistic):           4.17e-81
Time:                        09:51:16   Log-Likelihood:                 267.25
No. Observations:                 156   AIC:                            -512.5
Df Residuals:                     145   BIC:                            -479.0
Df Model:                          10
Covariance Type:                  HC3
==============================================================================================================
                                                coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------------------------
Intercept                                     0.2829      0.144      1.971      0.049       0.002       0.564
C(income_group)[T.Low income]                 0.0285      0.036      0.796      0.426      -0.042       0.099
C(income_group)[T.Lower middle income]        0.0065      0.026      0.249      0.803      -0.044       0.057
C(income_group)[T.Upper middle income]       -0.0183      0.017     -1.060      0.289      -0.052       0.016
hc                                            0.0479      0.010      4.715      0.000       0.028       0.068
Subscriptions                                 0.0021      0.001      3.812      0.000       0.001       0.003
log_GDP_emp                                  -0.0151      0.012     -1.244      0.213      -0.039       0.009
Internet                                      0.0009      0.000      2.452      0.014       0.000       0.002
log_Mobile                                    0.0203      0.015      1.355      0.176      -0.009       0.050
Q("Service share")                            0.0003      0.001      0.490      0.624      -0.001       0.001
Q("Government Effectiveness")                 0.0878      0.008     10.744      0.000       0.072       0.104
==============================================================================
```

Hierarchical Clustering

	Income group	Clusters (k)	Silhouette score
0	High income	2	0.2417
1	High income	3	0.2318
3	High income	5	0.2112
2	High income	4	0.2087
12	Low income	2	0.2949
15	Low income	5	0.1928
13	Low income	3	0.1761
14	Low income	4	0.1735
10	Lower middle income	4	0.1910
8	Lower middle income	2	0.1903
9	Lower middle income	3	0.1828
11	Lower middle income	5	0.1810
5	Upper middle income	3	0.1713
4	Upper middle income	2	0.1667





Ward's hierarchical clustering method: groups countries by minimising the increase in within-cluster variation as clusters are merged.

Hierarchical Clustering



- Group 1: Cluster 2 has a higher AI Index than Cluster 1, at 0.739 compared with 0.574. It also has higher human capital, broadband subscriptions, internet usage and government effectiveness.
- Group 2: Cluster 2 has the highest AI Index, at 0.513, Cluster 1 has a slightly lower AI Index of 0.446. Cluster 3 has the lowest AI Index, at 0.352.
- Group 3: Within lower-middle-income countries, Cluster 1 has a higher AI Index than Cluster 2, at 0.401 compared with 0.325.
- Group 4: Cluster 2 has a higher AI Index than Cluster 1, at 0.344 compared with 0.289.

Key Findings

- AI readiness varies within income groups
- Key factors associated with AI readiness: Human capital, fixed broadband subscriptions, internet use and government effectiveness.
- Hierarchical clustering identifies distinct nine country cluster profiles across the four income groups, reflecting different combinations of AI readiness, skills, digital connectivity, labour productivity and governance.
- Income alone does not capture country readiness
- Policy implications should reflect country profiles

Limitations and Future Scope

- Missing data required imputation, which may affect some estimates.
- The AI Index measures AI readiness, not actual AI adoption or realised productivity gains.
- Clustering results depend on the variables and number of clusters selected.
- Regression results show associations, not causal relationships.

- Use panel data to examine how AI readiness and productivity evolve over time.
- Incorporate direct measures of AI adoption and AI-driven productivity gains.
- Conduct country-specific studies to identify targeted policy interventions.

Conclusion

1 What we did

- Studied why AI readiness differs between countries in the same World Bank income group.
- Used 156 countries and the IMF AI Preparedness Index as the outcome.
- Linked it to human capital, digital access, productivity and governance.

2 How we did it

- Merged IMF, World Bank and Penn World Table data, then cleaned, imputed and log-transformed it.
- Explored it with box plots, scatter plots, correlations and VIF checks.
- Ran OLS regression with income-group controls ($R^2 = 0.915$).
- Used Ward hierarchical clustering inside each income group.

3 What we found

- AI readiness varies a lot within every income group.
- Government effectiveness, human capital, broadband subscriptions and internet use are significantly and positively linked to AI readiness.
- Strong pooled correlations mostly reflect rich-vs-poor gaps.
- Clustering gave 9 country profiles, so income alone does not capture readiness.

ThankYou

